

Product Data Architecture

EXEC SUMMARY

A composable digital platform is only as reliable as the data that flows through it. This document makes the architectural case for Pimcore as BSAVA's product master — the single source of truth from which all downstream systems (Contentful, Algolia, Commerce Layer, Salesforce) derive consistent, validated product data. Without it, the composable stack fractures at its most critical joint.

1. The Problem Statement

BSAVA's product catalogue is not a simple list of SKUs. It is a heterogeneous set of entities — physical books, digital resources, tiered memberships, CPD events, and online courses — each with distinct attributes, pricing logic, regulatory metadata, and entitlement rules. Managing this complexity across distributed spreadsheets or embedding it inside a headless CMS creates a class of architectural problems that compound over time.

1.1 The Distributed Spreadsheet Anti-Pattern

Prior to this migration, product data resided in a fragmented collection of Excel and Google Sheets files. From an architectural standpoint, this presents several systemic risks:

- Version drift — 'Final_Prices_v3_March.xlsx' is not a data store; it is a liability. There is no single version of truth, and reconciliation is a manual, error-prone process.
- No real-time API surface — a headless frontend cannot consume a spreadsheet without fragile, bespoke glue code that breaks silently.
- Asset-data separation — product images and metadata live in different locations with no enforced relationship between them. Pimcore's integrated DAM resolves this by co-locating digital assets with their product records.
- No validation layer — there is nothing preventing a missing ISBN, an invalid date range on an event, or a blank membership tier from propagating into the storefront.

1.2 Why the CMS Cannot Carry This Load

Contentful is an excellent choice for editorial content — clinical articles, campaign pages, UI copy, and structured marketing content. It is not, however, the right tool for product master data. The distinction matters architecturally:

KEY POINT

Contentful is optimised for the 'Who, What, and Why' (Storytelling)

Salesforce is for "Who are You?" (Identity & Entitlement).

Pimcore is optimised for the 'How Much, What Kind, and Under What Terms' of product specification.

*Conflating these concerns into a single system **creates** a monolith in composable clothing.*

Specific limitations of CMS-as-PIM for BSAVA's requirements:

- **Member vs. non-member pricing ('If user type = Vet, then price = £X') cannot be modelled as a content type without creating fragile workarounds. In Pimcore, memberPrice and nonMemberPrice are first-class attributes on the product schema.:** Non-editorial logic
- **Contentful is designed to publish content outward to a frontend. Pimcore is designed to syndicate structured data to multiple downstream consumers simultaneously — Contentful, Algolia, Commerce Layer, and Salesforce. The data flow directions are architecturally different.:** Syndication topology
- **A CPD event that bundles a digital resource, carries a membership discount, and requires attendance capacity management cannot be cleanly expressed in a CMS content model without significant schema contortion.:** Relationship complexity
- **Algolia's relevance is a direct function of the quality and structure of the data fed into it. Product data enriched and validated through a PIM produces materially better search results than product records cobbled together in a CMS.:** Search index quality

2. The Architectural Case for Pimcore

Pimcore functions as the gravity centre of the BSAVA product data mesh. It does not merely store products — it governs them. The following sections articulate the specific architectural roles Pimcore fulfils.

2.1 Complex Domain Modelling

BSAVA's product entities have requirements that demand a rigorous data modelling capability. Pimcore's PHP class definitions enforce these schemas at the platform level, preventing invalid records from being created in the first place.

01

Books — ISBN and Edition Integrity

Physical and digital publications require ISBNs, multi-author relationship graphs, edition tracking, and zero-rated VAT classification. A strict schema prevents incomplete or duplicate records from reaching Commerce Layer.

02

Memberships — Tiered Entitlement Logic

Vet, Nurse, and Student membership tiers are not merely price points; they are entitlement keys that unlock discounted pricing across the entire catalogue. This logic belongs in the PIM, where it can be consistently applied to every product, not duplicated across CMS entries.

03

CPD Events — Temporal and Capacity Attributes

Events require start/end datetimes, venue or virtual delivery modality, attendance capacity, and cancellation policy metadata. These are product specifications, not editorial content, and they must be kept in sync with Swoogo, the events management platform.

04

Online Courses — LMS Integration Metadata

Course records must carry Brightspace identifiers, prerequisite relationships, and CPD credit values. The PIM holds the canonical version of this metadata and syndicates it — it does not duplicate it from the LMS.

2.2 Pimcore as the Integration Mesh

In the MACH architecture, Pimcore occupies a unique position: it both consumes data from specialist services and syndicates enriched product records to consumers. This bidirectional role is what makes it the integration hub, not merely a data store.

Inbound data integrations (Pimcore as consumer):

- Swoogo → Pimcore: Live event capacity, ticketing status, and scheduling data flows into Pimcore to keep event product records current without manual intervention.
- Brightspace → Pimcore: Course metadata, module structure, and CPD credit values are synchronised from the LMS to ensure the storefront reflects the authoritative learning content.

Outbound data syndication (Pimcore as publisher):

- Pimcore → Algolia: Structured, taxonomy-driven product data is indexed directly into Algolia, ensuring search relevance and faceted navigation are driven by clean master data.
- Pimcore → Contentful: Product reference data is pushed to Contentful where editorial teams can create contextual content around it — without owning or duplicating the product record itself.
- Pimcore → Commerce Layer: Validated SKUs, pricing tiers, and tax category codes are the source of truth for the order management system. Commerce Layer does not decide what a product costs or how it is taxed; Pimcore does.
- Pimcore → Salesforce: Product and order metadata, including nominal codes for revenue recognition, flows into Salesforce for CRM and financial reporting.

NOTE

This hub-and-spoke topology is only sustainable because Pimcore is the authoritative source. If product data were distributed across the CMS, spreadsheets, or individual service databases, every integration would need to reconcile conflicting versions — the hallmark of a fragile, point-to-point architecture.

2.3 Pricing and Tax Architecture

Pricing logic is a domain boundary that must be allocated deliberately. In BSAVA's architecture, responsibility is split by capability:

- Tax category ownership (Pimcore): Whether a product is zero-rated (physical books under UK VAT rules) or standard-rated (events, digital goods) is a product attribute — it belongs in the PIM. Pimcore stores the TaxCategory code on each product record.
- Tax calculation (Commerce Layer / Stripe Tax): The actual VAT amount, based on the customer's billing country and the product's tax category, is computed at checkout. This separation ensures that pricing logic does not bleed into the product catalogue.
- Member pricing (Pimcore): The memberPrice and nonMemberPrice attributes are managed as first-class fields on every product. These values are passed to Algolia (for display) and Commerce Layer (for calculation), ensuring a single point of truth.

2.4 Legal and Compliance Metadata

Regulatory and compliance metadata requires the same careful domain allocation as pricing:

- Generic legal content (Contentful): Terms and conditions, privacy policies, and right-to-cancel notices are editorial content with a defined publication lifecycle. They belong in the CMS.
- SKU-specific warnings (Pimcore): 'Professional registration required' or 'Restricted to UK members only' are product specifications. They must be indexed and surfaced on the product detail page. They belong in the PIM.
- Financial compliance codes (Pimcore → Salesforce): Nominal codes for revenue recognition — particularly relevant for 12-month membership subscriptions where income must be

deferred — are stored as Pimcore product attributes and flow to Salesforce on the order record.

3. Capability Comparison

The table below summarises the architectural fit of each approach against BSAVA's requirements:

Capability	Spreadsheets	Headless CMS (Contentful)	PIM / MDM (Pimcore)
Primary Purpose	Ad-hoc data entry	Editorial storytelling & pages	Structured product lifecycle mgmt
Data Integrity	Poor — version drift, typos	Moderate — basic field types	High — strict schema validation
Relationship Modelling	Manual / fragile	Simple references	Complex bundles, variants, CRM sync
Scalability	Low — file-based	High — API-first	High — DB-driven, mass-edit capable
Channel Syndication	None	Limited — publish to web	Multi-channel: web, search, commerce
BSAVA Use Case	Legacy data migration	News, clinical articles, UI copy	Books, memberships, CPD events

4. Design Principles

This architecture is grounded in four principles that should govern all future data modelling decisions:

P1

Single Source of Truth

Every product attribute has one authoritative owner. Duplication across systems is a bug, not a feature. When Swoogo updates an event's capacity, that change flows through Pimcore to all consumers — no manual re-entry.

P2

Strict Schema Enforcement

Pimcore's PHP class definitions are the contract. A product record that does not satisfy the schema does not enter the system. This shifts data quality from a QA activity to a platform constraint.

P3

Separation of Concerns

Contentful owns storytelling. Pimcore owns product specification. Salesforce owns identity and entitlement. Commerce Layer owns transactional logic. These boundaries must be maintained as the platform evolves — resist the temptation to add product logic to the CMS or editorial content to the PIM.

P4**API-First Syndication**

All data flows are API-driven. No manual exports, no scheduled file transfers, no spreadsheet imports. Every integration is a first-class citizen of the architecture with defined contracts, error handling, and monitoring.

5. Conclusion

The composable architecture represented in BSAVA's target state diagram is only as strong as the governance of the data flowing through it. A specialist PIM is not an optional enhancement — it is the structural element that makes composability coherent.

Without Pimcore as the product master, the architecture degrades into a set of powerful but disconnected tools, each holding a partial and potentially conflicting view of the product catalogue. With it, every downstream system — search, commerce, CMS, CRM — operates from a single, validated, governed version of the truth.

RECOMMENDATION

Pimcore should be treated as a platform capability, not a content tool. Its schema definitions, integration contracts, and syndication pipelines are engineering assets that require the same rigour, versioning, and governance as application code. Data architecture is system architecture.